Chih-Wei Joshua Liu

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EDUCATION

Stanford University — Stanford, CA, USA
Biophysics PhD Program · **Thesis advisor:** Grant M. Rotskoff · **GPA:** 4.0/4.0 — 09/2024 – present
Courses: IMMUNOL201 Advanced Immunology, SBIO242 Molecular Biophysics Methods, CS229 Machine Learning

Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA
Bachelor of Science · **Major:** Physics · **GPA:** 4.9/5.0 · **MCAT:** 528/528 — 05/2022
Courses: 18.650 Statistics, 8.591 Systems Biology · **Honors:** Joel Matthew Orloff Award, MIT News profile

National Experimental High School — Hsinchu, Taiwan, ROC
USABO: Semifinalist (2015 – 2018); High Honorist (2017) · **ACT:** 36/36 · **SAT:** 1600/1600 — 05/2018

EXPERIENCE

MIT Physics of Living Systems Group — 06/2022 – 08/2024
Research Assistant — Cambridge, MA, USA
Implemented microscopy experiments. Designed ensemble-learning and polyspectral analyses of biological images.

MIT Department of Physics — 02/2024 – 05/2024
Teaching Assistant, 8.241 Introduction to Biological Physics — Cambridge, MA, USA
Led recitations on protein folding, diffusion, and imaging methods. Assisted optical-trap experiments.

Massachusetts General Hospital — 06/2021 – 08/2021
Data Science Intern — Boston, MA, USA
Implemented GLS models of patient data and authored two clinical publications using `R`, `SPSS`, and `statsmodels`.

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, **Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data**, *Phys. Rev. X*, provisionally accepted (2026). [code]

- Derived the first maximum-likelihood parameterization of non-Markovian kinetics satisfying all physical constraints in general biomolecules. **Skills:** convex optimization, `lightning`
- Validated algorithm on CFTR ion-channel and F1 molecular-motor experiments. Set SOTA in benchmark extraction of HP35 protein-folding kinetics from molecular-dynamics simulations. **Skills:** autoregressive models, `mdtraj`

J. Li*, **C.-W.J. Liu***, M. Szurek, and N. Fakhri, **Measuring irreversibility from learned representations of biological patterns**, *PRX Life* **2**, 033013 (2024), *equal contribution. [code]

- Quantified energetic costs of actomyosin regulation by conserved Rho GTPase using only variational-autoencoder representations of cortical-wave microscopy data. **Skills:** `pillow`, `torch`, `torchvision`

A. Yona, **C.-W.J. Liu**, et al., and J. Gore, **Adaptive evolution identifies candidate genetic routes associated with oxalate utilization in probiotic and gut-derived *E. coli***, *Cell Rep. Methods* **7**, 101644 (2026).

- Led biochemical analysis of human gut microbes subject to directed evolution, characterizing loci repeatedly mutated across lineages with sequencing and imaging assays. **Skills:** sequence analysis, anaerobic technique, `pymol`

H. Lee*, E. Koskelo*, S. Gokhale*, J. Li*, C. Fei, **C.-W.J. Liu**, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, **Topological flowscape reveals state transitions in nonreciprocal living matter**, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution. Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

- Trained classifier of developmental stages in embryo videos, enabling identification of physically exotic swarming dynamics in multicellular aggregates. **Skills:** `scikit-learn`, `scipy`, `numpy`, `pandas`, `opencv`

GRANTS AND LEADERSHIP

National Science Foundation Graduate Research Fellowship (GRFP, 2024, Physics)
National Defense Science and Engineering Graduate Fellowship (2024, Physics, declined for GRFP)
MIT Society of Physics Students and Sigma Pi Sigma: honor-society inductee (2022); President (2022)
Phi Kappa Theta – Massachusetts Eta: Pledgemaster (2021); sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage as Service Chair (2019–2020).

SKILLS AND INTERESTS

Laboratory: cloning, enzyme assays, fluorescence microscopy · **Computing:** `pydantic`, `jax`, `uv`, `ty`, `ruff`, `slurm`
Natural language: Standard Chinese (native) · **Personal interests:** phonology, archaeology, ichthyology, Set
Research interests: nonequilibrium thermodynamics, biochemical kinetics, optimal transport, SAT solvers